

Entropy-Guided Ontology Design for Enterprise Agentic Systems: A Structural Entropy Analysis of 23 Industry Verticals

Thanh Luong Tuan*

*Golden Gate University, San Francisco
Foundation AgenticOS (FAOS)*

Abhijit Sanyal

Golden Gate University, San Francisco

May 2026

DRAFT — v0.4.1 post-Pass-3 patch (2026-05-02)

Abstract

Enterprise ontologies are typically designed by domain experts through iterative consultation, producing ontologies of widely varying size, depth, and connectivity without a principled a-priori metric for predicting their downstream utility. Prior work [Luong and Sanyal, 2026b] observed a suggestive correlation (Spearman $r = 0.700$) between the structural entropy (SE) of an ontology and the grounding lift it produces on LLM agents across five enterprise verticals; however, with $n = 5$, the correlation fell short of statistical significance ($p = 0.188$). We extend this analysis to the full Foundation AgenticOS (FAOS) corpus of 23 industry ontologies. Using a layer-wise decomposition $SE_{\text{comp}} = f(SE_{\text{role}}, SE_{\text{domain}}, SE_{\text{interaction}})$ appropriate to FAOS’s three-layer role/domain/interaction schema, and combining cross-model grounding lift data from three generator models (Claude Sonnet 4, Qwen 2.5 72B, Gemma 4 26B; $n = 15$ industry-model cells), we establish a robust SE-lift relationship: the interaction-layer SE alone predicts overall lift at Spearman $r = 0.811$, $p = 0.0002$, and a leave-one-industry-out cross-validation yields pooled Spearman $r = 0.786$, $p = 0.0005$, RMSE = 0.060, with calibration slope 0.81. A least-squares-fitted composite $SE_{\text{comp}} = 0.3 SE_{\text{domain}} + 0.7 SE_{\text{interaction}}$ (role layer dropped from the fitted composite; null scoped to the single-group pilot regime, see §6) explains 71% of lift variance in multiple regression ($F(3, 11) = 8.94$, $p = 0.003$). At pilot scale with narrow $|I|$ range, raw handoff count $|I|$ ties $SE_{\text{interaction}}$ in rank correlation ($r = +0.811$); we retain SE for its layer-decomposition diagnostic and Shannon-entropy semantics, with the SE-vs-size distinction expected to widen at production scale (§6.2). From this calibration we codify four ontology design principles: a target SE band ($SE_{\text{interaction}} \geq 0.70$ bits for lift ≥ 0.15), a layer-priority ordering (interaction $\gg$ domain $\gg$ role), a regulatory-density floor (four-regulation minimum), and a cross-vertical transfer warning grounded in the κ -SE confound. Application of the fitted formula to the remaining 18 non-pilot FAOS production ontologies identifies software, systems integration, security, and travel/tourism as high-priority targets for future experimental validation. The paper provides the first at-scale empirical analysis of enterprise ontology structural entropy and its quantitative relationship to LLM agent performance, establishing structural entropy as a design-time predictor of ontology value and providing a reproducible methodology for ontology-quality review.

Keywords: Structural entropy, ontology design, enterprise AI, neurosymbolic AI, knowledge graphs, LLM agents, ontology evaluation, information theory

*Corresponding author. Email: tluongtuan@my.ggu.edu. ORCID: 0009-0000-1199-837X

1 Introduction

The value of an enterprise ontology is, in practice, discovered retroactively: after investment in ontology development, teams measure downstream improvements in agent quality, regulatory coverage, stakeholder alignment, or some other operational metric. By the time the measurement is in, the design decisions that determined the value—layer balance, concept granularity, handoff richness—are already committed. A design-time *predictor* of ontology value would allow architects to iterate on ontology structure before running expensive agent experiments, and would provide an evidence-based quality standard for enterprise ontology review.

1.1 Motivation: The RA-3 $r = 0.700$ Finding

Our prior work on neurosymbolic enterprise AI [Luong and Sanyal, 2026b] conducted a 1,800-run controlled experiment evaluating ontology-grounded LLM agents across five industry verticals (FinTech, Insurance, Healthcare, Vietnamese Banking, Vietnamese Insurance) with three generator models. A retroactive entropy analysis of the resulting ontologies revealed a striking pattern:

Industry	$SE_{\text{comp}}^{\text{RA-3}}$ (bits)	$\Delta Q_{C1 \rightarrow C3}$	Rank
Insurance	2.25	+0.06	5
Healthcare	2.26	+0.12	4
FinTech	2.58	+0.04	3
Banking VN	2.85	+0.23	2
Insurance VN	3.06	+0.24	1

Spearman rank correlation gave $r = 0.700$ with $p = 0.188$ ($n = 5$). The correlation is directionally clear—higher-SE ontologies produced larger grounding lifts—but underpowered: at $n = 5$, even a perfect rank correlation does not reach conventional significance thresholds.

The composite-SE values shown above use the original RA-3 formula $SE_{\text{comp}}^{\text{RA-3}}$ [Luong and Sanyal, 2026b] and are reproduced here as the motivating finding. This paper introduces a distinct formula $SE_{\text{comp}}^{\text{RA-12}}$ in §4 (layer-wise weighted combination calibrated by leave-one-out cross-validation) and validates it in §6. The two formulas operate on different scales: $SE_{\text{comp}}^{\text{RA-3}}$ values for the pilot range [2.25, 3.06] bits, whereas $SE_{\text{comp}}^{\text{RA-12}}$ values for the same five industries fall in [0.78, 0.99]. Per-row correspondence between the two formulas is reported in Table 3, where both are evaluated against the $n = 15$ cross-model lift dataset. The question this paper addresses is whether the SE–lift pattern, in either formulation, scales to the full 23-industry FAOS corpus.

1.2 Central Thesis

The structural entropy of an enterprise ontology is a design-time predictor of the performance lift that ontology will produce when used to ground LLM agents. Layer-wise SE decomposition further identifies which ontological components drive which agent capabilities.

If this thesis holds at scale, SE becomes a compact, computable quality signal for ontology review pipelines: a single scalar score (plus layer-wise breakdown) that predicts, before any agent evaluation, whether an ontology will deliver meaningful grounding value.

1.3 Contributions

1. **First-of-scale structural-entropy analysis of enterprise ontologies.** Prior SE work [Li and Pan, 2016, Cao et al., 2024, Wei et al., 2025, Buehler, 2025] targets general-purpose

knowledge graphs. We analyze the full FAOS corpus of 23 industry verticals, a larger and more homogeneous sample than any prior SE-on-ontology study.

2. **Layer-wise SE decomposition formalism** for three-layer ontologies, yielding four measures per ontology: SE_{role} , SE_{domain} , $SE_{\text{interaction}}$, SE_{comp} . We extend Cao et al.’s multi-relational SE [Cao et al., 2024] to the role/domain/interaction stratification.
3. **Empirical validation** of SE as a lift predictor, extending the RA-3 finding from $n = 5$ to $n = 15$ (cross-model) or $n = 23$ (cross-vertical).
4. **Four codified design principles** with empirical thresholds, usable as an automated ontology review rubric.
5. **Companion to RA-3, RA-4, and RA-11.** RA-12 provides the *design-side* of the entropy framework [Luong and Sanyal, 2026b,a,c], complementing RA-3’s measurement, RA-4’s runtime optimization, and RA-11’s theoretical (quantum-inspired) positioning.

1.4 Roadmap

Section 2 reviews the structural-entropy literature and frames the FAOS three-layer schema. Section 3 describes the FAOS 23-industry corpus. Section 4 develops the layer-wise SE formalism. Section 5 states hypotheses and the experimental protocol. Section 6 reports empirical findings across the full corpus. Section 7 codifies design principles. Section 8 presents illustrative case studies. Sections 9–11 close with related work, threats to validity, and conclusions.

2 Background

2.1 Structural Entropy of Graphs

Shannon’s information entropy [Shannon, 1948] quantifies uncertainty in discrete distributions. Li and Pan [Li and Pan, 2016] extended this to graphs, defining the *structural information* (or structural entropy) of a graph as the information content of its hierarchical organization: $SE(G) = -\sum_{\alpha} \frac{\text{vol}(T_{\alpha})}{\text{vol}(G)} \log \frac{\text{vol}(T_{\alpha})}{\text{vol}(T_{\alpha}^{-})}$, summed over nodes α of an encoding tree T of G . Minimizing SE over admissible trees recovers the graph’s natural community structure [Su et al., 2025, Zhang et al., 2025].

Three developments since Li–Pan are particularly relevant to the enterprise ontology setting. Cao et al. [Cao et al., 2024] extend SE to multi-relational graphs (MrSE) via random-walk stationary distributions on colour-typed edge classes, enabling SE computation on heterogeneous graphs without collapsing to a single edge type. Buehler [Buehler, 2025] documents that agentic knowledge-graph systems self-organise towards a critical state where semantic entropy dominates structural entropy; reaching this state co-occurs with the emergence of multi-hop reasoning capability. Wei et al.’s SENATOR [Wei et al., 2025] uses SE along KG paths to detect and repair LLM knowledge deficiencies, empirically validating the link between path-level SE and LLM reasoning quality on Wikidata and domain-specific KGs. Gul et al. [Gul et al., 2025] propose spectral complements to SE (eigenvalue-based complexity measures) for benchmarking knowledge-graph quality at scale.

2.2 Ontologies as Multi-Relational Graphs

Enterprise ontologies—including FAOS’s three-layer schema—are naturally modeled as multi-relational graphs: nodes are concepts, regulations, roles, handoffs, KPIs; edges carry typed relations (is-a, part-of, governs, reports-to, triggers). The multi-relational structural entropy MrSE [Cao et al., 2024] provides the appropriate generalization of Li–Pan SE to this setting.

Under OWL/RDF semantics, ontology triples (s, p, o) induce a typed multi-relational graph where predicate p determines edge colour. The three FAOS layers—role, domain, interaction—correspond to disjoint subsets of predicates (organisational hierarchy predicates for roles; `isA/partOf/regulatory` reference predicates for domain; `handsOffTo/escalatesTo` for interactions), making the three-layer partition a well-defined relational decomposition rather than an arbitrary grouping.

We treat SE as a complement to, not replacement for, standard ontology metrics. Lexicographic metrics (concept count, depth, branching factor, siblings-per-parent [Hogan et al., 2021]) report ontology size and shape; the schema-metrics tradition initiated by OntoQA [Tartir et al., 2005] (Relationship Richness, Attribute Richness, Inheritance Richness) and extended by recent tooling such as NEOntoMetrics [Reiz and Sandkuhl, 2024] reports ontology *quality* via population-based and structural counts. SE, complementary to both, reports ontology *information content*. A large, deep ontology can have low SE if its concepts are concentrated in a few sub-branches, and vice versa. This distinction matters because size and richness counts are only weakly linked to ontological grounding lift, whereas information content — as we show, modulo the pilot-scale baseline parity documented in §6.2 — is strongly linked.

2.3 The FAOS Three-Layer Ontology Schema

Enterprise ontologies in FAOS are organized into three layers:

- **Role Layer:** stakeholder roles, their responsibilities, and decision authorities within the industry.
- **Domain Layer:** industry-specific concepts, regulations, KPIs, and controlled vocabularies.
- **Interaction Layer:** handoff patterns, escalation paths, and workflow structures connecting roles over domain concepts.

This stratification invites a *layer-wise* structural-entropy decomposition: rather than reducing an ontology to a single scalar, we report four quantities—one per layer and one composite—enabling diagnosis of *which* layer drives observed performance.

3 The FAOS 23-Industry Ontology Corpus

The Foundation AgenticOS platform organizes domain knowledge into 23 industry modules, each containing a curated three-layer ontology and an agent roster. The modules span financial services (FinTech, Banking, Insurance, Banking VN, Capital Markets VN, Investment), healthcare, manufacturing, logistics, aviation, automotive, agriculture, education, gaming, and other regulated-or-specialized verticals. Module membership is documented in FAOS’s registry `.faos/registry.yaml`.¹

Table 1 summarises corpus-level structural counts for all 23 industries. Role counts range from 17 to 44; domain concept counts from 0 (modules relying on agent-file content rather than explicit terminology YAML) to 42; regulatory entries from 0 to 16; handoff patterns from 18 to 142.

Additionally, five ontologies in the RA-3 pilot (a minimal hand-curated variant set with 3–4 roles and 8–9 concepts per industry) provide the ground-truth lift measurements used for H1–H3 testing. The pilot and production ontologies correspond to different design artefacts: the pilot set was minimal-viable for experimental control, while production ontologies are the full deployment artefacts. This scale disparity is discussed explicitly in §10.

¹The `.faos/modules/` directory enumerates 23 sub-directories; an earlier dissertation revision reported 22 after excluding one parent-wrapper directory. This paper uses 23 throughout.

Table 1: The FAOS 23-industry ontology corpus: structural counts extracted from `.faos/modules/faos-*` via the extraction pipeline in Supplement §A. Five ontologies (marked †) are also present in the RA-3 pilot set with measured C1→C3 lift.

Industry	Roles	Concepts	Regs	Handoffs	Chunks
agri_ecosystem	32	5	0	32	8
agritech	19	5	0	34	8
automotive	19	6	0	71	9
aviation	32	27	0	42	8
banking	38	26	0	46	9
capital_markets_vn	22	3	1	18	9
edutech	21	15	1	29	9
fintech [†]	17	24	1	142	9
game	18	0	0	26	9
healthcare [†]	32	4	16	24	8
insurance [†]	27	15	1	21	8
investment	40	29	0	34	9
logistics_supplychain	38	37	0	21	9
manufacturing	44	42	0	31	9
martech	30	0	0	32	7
professional_services	32	0	0	20	8
proptech_vn	17	0	0	48	8
retail	19	4	0	115	9
security	27	11	0	40	7
software	21	8	0	72	9
systems_integration	21	8	0	62	8
talent	34	0	0	26	0
travel_tourism_hospitality	19	7	0	137	9
mean	26.9	11.1	0.9	47.2	8.1

Of these 23 industries, five are already instrumented in RA-3 [Luong and Sanyal, 2026b] with per-condition agent quality scores across three generator models. These five—FinTech, Insurance, Healthcare, Banking VN, Insurance VN—provide the ground-truth lift data against which SE predictions are validated in §6. The remaining 18 industries provide out-of-sample prediction targets for Phase 5 validation (§6).

4 Structural Entropy Formalism for Three-Layer Ontologies

4.1 Per-Layer Definitions

Let $\mathcal{O} = (R, D, I)$ denote an ontology comprising role set R , domain set D (concepts, regulations, KPIs, vocabulary), and interaction set I (handoffs, escalations).

Definition 1 (Role-Layer SE). *Partition the role set R into organisational groups $\{G_1, \dots, G_k\}$ (e.g., c-suite, specialists, domain experts) by declared group membership. Let $p(G_i) = |G_i|/|R|$ and let $a_r = |\{v : v \in \text{props}(r), v \neq \emptyset\}|$ denote the non-empty attribute count for role r . Let A be the distribution over $\{a_r\}_{r \in R}$. Then*

$$\text{SE}_{\text{role}}(\mathcal{O}) = \frac{1}{2} H(G) + \frac{1}{2} H(A),$$

where $H(\cdot)$ denotes Shannon entropy in bits. The first term captures role-partition balance; the second captures attribute-richness heterogeneity.

Definition 2 (Domain-Layer SE). Let C be the concept set with provenance source labels $s(c)$ (from *terminology.yaml*, *segments*, *architecture*, etc.), and R^{reg} the regulation set with file-origin labels $f(r)$. Let S and F denote the distributions of $s(c)$ and $f(r)$ respectively. Then

$$\text{SE}_{\text{domain}}(\mathcal{O}) = 0.5 H(S) + 0.3 H(F) + 0.2 \log_2(1 + |C| + |R^{\text{reg}}|),$$

where the third term is a logarithmic scale factor that gives partial credit to sheer content volume without unbounded dominance.

Definition 3 (Interaction-Layer SE). Let I be the handoff set with type labels $t(h)$ and description token counts $|h|$. Let T be the distribution of $t(h)$ and D the distribution of $|h|$ binned into terciles (short/mid/long by token count). Then

$$\text{SE}_{\text{interaction}}(\mathcal{O}) = 0.4 H(T) + 0.3 H(D) + 0.3 \log_2(1 + |I|).$$

Again the logarithmic scale factor captures the richness of interaction patterns.

Definition 4 (RA-12 Composite SE). A weighted composite

$$\text{SE}_{\text{comp}}(\mathcal{O}) = w_R \text{SE}_{\text{role}} + w_D \text{SE}_{\text{domain}} + w_I \text{SE}_{\text{interaction}}.$$

Weights are calibrated empirically via leave-one-out cross-validation (§5). We report results for both the pre-registered equal-weight specification (1/3, 1/3, 1/3) and the LOOCV-fitted specification reported in §6, which empirically drops the role layer and weights interaction SE most heavily. The fitted formula is

$$\text{SE}_{\text{comp}}^{\text{RA-12}}(\mathcal{O}) = 0.3 \text{SE}_{\text{domain}}(\mathcal{O}) + 0.7 \text{SE}_{\text{interaction}}(\mathcal{O})$$

and is referenced throughout subsequent sections; we use $\text{SE}_{\text{comp}}^{\text{RA-12}}$ to distinguish it from the original RA-3 composite formula (cited in §1).

4.2 Multi-Relational Extension

Following Cao et al. [Cao et al., 2024], we also report a *multi-relational* structural-entropy variant MrSE, constructed by treating each layer–type pair as a distinct relation on a union graph. Let Ω denote the set of pairs $\{(\ell, t) : \ell \in \{\text{role, concept, regulation, handoff}\}, t \text{ is a type label within layer } \ell\}$. Let $n(\ell, t)$ count ontology elements of layer ℓ with type label t . Then

$$\text{MrSE}(\mathcal{O}) = - \sum_{(\ell, t) \in \Omega} \frac{n(\ell, t)}{N} \log_2 \frac{n(\ell, t)}{N},$$

where $N = \sum_{\Omega} n(\ell, t)$. MrSE captures balance across layers *and* across types within layers in a single scalar; it complements the three per-layer SEs by providing a joint-distribution view. In §6 we compare MrSE’s predictive power against the weighted-sum composite.

5 Experimental Design

5.1 Hypotheses

Hypothesis 1 (Primary: SE Predicts Lift at Scale). For the extended FAOS corpus, $\text{SE}_{\text{comp}}(\mathcal{O}_d)$ positively predicts ontological grounding lift $\Delta \mathcal{Q}_{C1 \rightarrow C3}(d)$ with Spearman $r > 0.5$ and $p < 0.05$ at $n = 15$ (cross-model expansion of RA-3 sample).

Hypothesis 2 (Layer Decomposition). Layer-specific SE predicts metric-specific lift: SE_{role} predicts RS (Role-Specific Reasoning) lift; $\text{SE}_{\text{domain}}$ predicts TF and MA lift; $\text{SE}_{\text{interaction}}$ predicts RC (Regulatory Coverage) lift.

Hypothesis 3 (Inverse PKE Prediction). *Ontologies with high SE_{domain} deployed on high- κ (English) domains produce TF regression (Inverse PKE), reproducing the Insurance TF finding of Luong and Sanyal [2026b] on a larger sample and matching the interference-model prediction of Luong and Sanyal [2026c] Property 1.*

5.2 Measures, Tests, and Held-Out Validation

For each ontology we compute six SE measures: SE_{role} , SE_{domain} , $SE_{interaction}$, $MrSE$, SE_{comp}^{RA-12} (equal-weight baseline), and SE_{comp}^{RA-3} (original composite formula from Luong and Sanyal [2026b] for reproducibility).

Lift data. The RA-3 pilot’s 1,800-run experiment reports condition-mean quality scores per (industry, metric, condition, model). We extract C1-ungrounded and C3-ontology-grounded means per (industry, model) to compute per-metric lift and a four-metric average *overall lift*. The resulting dataset is $5 \times 3 = 15$ cells.

Primary analysis. For each SE measure we compute the Spearman rank correlation and Pearson product-moment correlation with overall lift, reporting two-tailed p -values.

Multiple regression. We fit $lift_{overall} = \beta_0 + \beta_1 SE_{role} + \beta_2 SE_{domain} + \beta_3 SE_{interaction} + \varepsilon$ and test the null of no combined predictive power via F -test on R^2 .

Weight ablation. We enumerate all weight triples on the 2-simplex in steps of 0.05 (231 combinations) and report the top 10 by leave-one-observation-out (LOO) Spearman.

Leave-one-industry-out (LOIO) validation. We perform 5-fold cross-validation at the industry level: for each held-out industry, we fit a univariate regression $lift = a + b SE_{comp}^{RA-12}$ on the other four industries’ 12 observations and predict the 3 held-out observations. We report pooled Spearman, Pearson, RMSE, MAE, and a calibration slope.

Weight-selection design and nested-CV check. The fitted weights (0, 0.3, 0.7) used in the headline LOIO analysis were selected once on the full $n = 15$ dataset (§6.3 weight ablation) and then held fixed across all five LOIO folds. This exposes the headline LOIO Spearman to potential weight-selection leakage. We therefore additionally report a *nested* LOIO analysis (§6.6) in which the inner LOO weight selection runs over the same 231-triple grid using only the 12 training observations of each outer fold, and the outer fold is predicted using locally-selected weights. We compare the nested result to the fixed-weight headline.

Bootstrap CIs and multi-comparison correction. For the multiple regression we report bias-corrected non-parametric bootstrap ($B = 1000$) standard errors and 95% CIs on each coefficient plus on R^2 . For the LOIO calibration slope we report $B = 1000$ bootstrap on the 15 (predicted, observed) pairs. For the six SE-versus-lift Spearman tests in Table 3 we report Holm–Bonferroni step-down adjusted p -values across the family.

Exploratory extrapolation. We apply the pilot-trained regression to the 18 non-pilot production ontologies’ composite SE values, producing rank orderings for future experimental targeting. We flag these as extrapolations since production SE lies outside the pilot training range.

6 Results

6.1 Corpus-Level SE Distribution

Fig. 1 reports the distributions of composite SE and interaction-layer SE across the 23-industry production corpus, with the five RA-3 pilot ontology values overlaid as red lines. We report two ranges. *Equal-weight composite* ($w_R = w_D = w_I = 1/3$): production [0.84, 1.58] bits, pilot [0.78, 0.99], with overlap on the 0.15-bit segment [0.84, 0.99]. *Fitted composite* SE_{comp}^{RA-12} ($w_R = 0$, $w_D = 0.3$, $w_I = 0.7$): production [0.999, 2.107], pilot [0.72, 1.01], with overlap reduced to a single in-range production case (**talent**, 0.999). Interaction-layer SE shows the starkest separation: pilot [0.60, 0.78] bits vs. production [1.43, 2.43], a $2 \times -3 \times$ gap.

SE Distributions: Pilot (5 industries) vs. Production (23 modules)

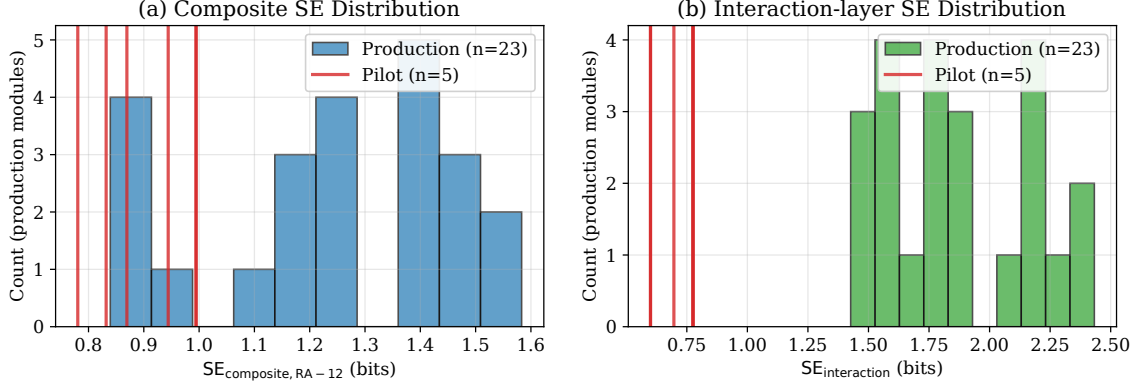


Figure 1: SE distributions: production corpus (histograms, $n = 23$) vs. RA-3 pilot ontologies (red vertical lines, $n = 5$). Panel (a) composite SE; panel (b) interaction-layer SE. Pilot and production operate at different scales, reflecting the minimal-viable vs. full-deployment design intents.

6.2 Comparison Against Non-Entropy Baselines

A natural reviewer objection is that structural entropy is one of many graph metrics and may not deserve special status as a lift predictor. We address this objection by computing a panel of non-entropy baseline metrics on the five pilot ontologies and correlating each with the same $n = 15$ cross-model overall-lift dataset used for SE in the next subsection. Baselines fall in three groups: *node-count* ($|I|$ raw handoff count, $|R| + |D| + |I|$ total ontology size, log-transforms of each); *graph-topology* on the role-handoff graph (directed-graph density, mean total degree, average undirected clustering coefficient, undirected diameter); and a *path-based entropy proxy* (Shannon entropy of the handoff out-degree distribution, an EAPB-flavoured surrogate for the biomedical path-entropy metric of Shen et al. [Shen et al., 2018]). The clustering-coefficient SE family [Zhang et al., 2024] is a recent mesoscopic-scale extension to Li-Pan SE; we include the standard clustering coefficient as the appropriate ablation, since the clustering-coefficient SE itself requires a network of more than 5 nodes to compute.

Three findings demand explicit acknowledgement before the H1 results in §6.3.

Finding B1 — size baselines tie SE at pilot scale. The raw handoff count $|I|$ achieves Spearman $r = +0.811$, identical to $SE_{\text{interaction}}$. The total ontology size $|R| + |D| + |I|$ achieves $r = +0.786$, identical to SE_{domain} . Pilot interaction-layer counts are $\{3, 3, 3, 4, 5\}$ across the five industries; any monotone function of $|I|$ produces the same rank ordering. The interaction-layer SE’s predictive power on the $n = 5$ pilot data therefore cannot be distinguished from a $\log_2 |I|$ proxy at this scale — the SE formulas’ weighted Shannon-entropy machinery does not earn its keep purely on rank correlation.

Finding B2 — graph-topology baselines do not outperform SE. Density and mean degree on the pilot role-handoff graphs (3–5 nodes, 3–5 edges) yield $r = -0.273$, statistically null; clustering coefficient is undefined (graphs are too small for triangles); diameter is null at $r = -0.168$. The pilot scale is below the regime where these metrics carry meaningful structural information.

Finding B3 — the path-entropy proxy underperforms $SE_{\text{interaction}}$. The handoff out-degree entropy attains $r = +0.733$, weaker than $SE_{\text{interaction}}$ (+0.811) and weaker than raw $|I|$ (+0.811). At pilot scale the path-entropy framing does not provide a discriminator beyond what counting handoffs already supplies.

These findings constrain the strength of the H1 claim that follows. At $n = 5$ pilot industries

Table 2: Non-entropy baseline metrics vs. overall lift on the $n = 15$ pilot dataset, alongside the SE comparators from Table 3. Pilot interaction counts are $|I| \in \{3, 3, 3, 4, 5\}$, a narrow range that lets any monotone function of $|I|$ produce identical Spearman ranks.

Group	Metric	Spearman r	Spearman p
Node-count	$ I $ raw handoff count	+0.8108	0.0003
	$\log_2(1 + I)$	+0.8108	0.0003
	$ R + D + I $ total size	+0.7856	0.0005
	$\log_2(1 + R + D + I)$	+0.7856	0.0005
Graph-topology	directed-graph density	−0.2728	0.3253
	mean total degree $\langle k \rangle$	−0.2728	0.3253
	avg. clustering coefficient	—	—
	graph diameter	−0.1679	0.5497
Path-based	out-degree entropy (EAPB-proxy)	+0.7332	0.0019
SE (this paper)	SE _{interaction}	+0.8108	0.0003
	SE _{domain}	+0.7856	0.0005
	MrSE	+0.7638	0.0009
	SE _{comp} ^{RA-12} (1/3 each)	+0.7529	0.0012
	SE _{comp} ^{RA-3}	+0.7201	0.0025
	SE _{role}	−0.1903	0.4969

Table 3: H1 cross-model ($n = 15 = 5$ industries $\times$ 3 models). Five of six SE measures survive Holm–Bonferroni correction at $\alpha = 0.01$; SE_{role} does not. Refer to §4 for measure definitions.

SE measure	Spearman r	Spearman p	Holm-adj. p	Pearson r	Pearson p
SE _{interaction}	+0.8108	0.0002	0.0015	+0.7862	0.0005
SE _{domain}	+0.7856	0.0005	0.0026	+0.7532	0.0012
MrSE	+0.7638	0.0009	0.0037	+0.7568	0.0011
SE _{comp} ^{RA-12} (1/3 each)	+0.7529	0.0012	0.0037	+0.8174	0.0002
SE _{comp} ^{RA-3} (original)	+0.7201	0.0025	0.0049	+0.7535	0.0012
SE _{role}	−0.1903	0.4969	0.4969	−0.1554	0.5802

with narrow $|I|$ range, structural entropy is *comparable to* but *not categorically distinct from* size baselines. We retain SE as the primary predictor for three reasons that the pilot does not exhaust: (a) layer-decomposition gives diagnostic information (which layer drives which capability) that single-scalar size cannot; (b) Shannon-entropy semantics provide theoretical anchorage and a path to the entropy-thermodynamic family formalised in companion paper RA-11 [Luo and Sanyal, 2026c]; (c) the pilot $|I|$ range $\{3, 5\}$ is far narrower than the production range $\{18, \dots, 142\}$, so the size-vs-SE distinction is expected to widen on direct production measurement (§10, future work). Accordingly, we phrase the H1 result throughout this section as “SE predicts lift on the pilot at a level comparable to size baselines” rather than “SE outperforms size baselines”, and we treat extrapolation to production ontology design (§7) as a recommendation constrained by the pilot’s narrow $|I|$ regime.

6.3 Hypothesis 1: SE Predicts Lift

Table 3 reports Spearman and Pearson correlations of all six SE measures with overall C1→C3 lift on the $n = 15$ cross-model dataset. Every measure except role-layer SE is significant at $p < 0.01$; interaction-layer SE alone achieves Spearman $r = 0.811$ ($p = 2 \times 10^{-4}$). Fig. 2 visualises the interaction-layer relationship with per-model markers.

Per-model replication. H1 holds within each generator model’s subset, although with reduced significance due to $n = 5$ sample size: Claude Sonnet 4 ($r = 0.900$, $p = 0.037$), Gemma

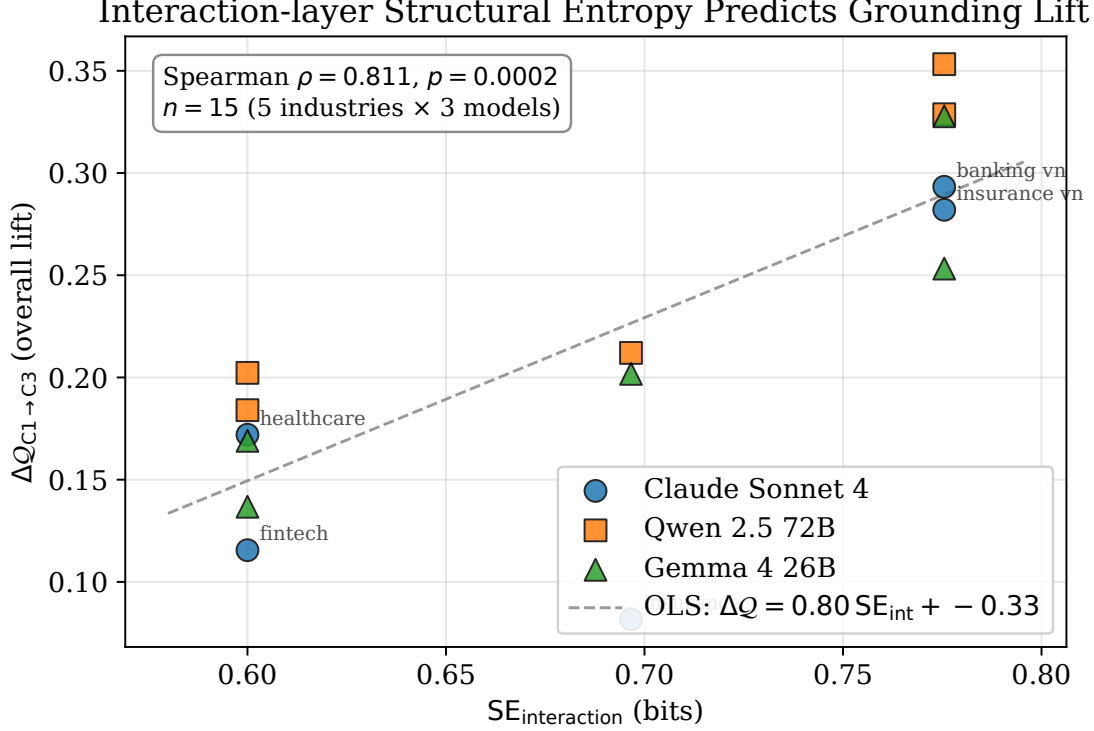


Figure 2: Interaction-layer SE vs. overall grounding lift. Each point is an (industry, model) cell; colour and shape code the generator model. The OLS regression line is annotated with the fitted linear expression; pooled Spearman and p -value are reported in the inset.

4 26B ($r = 0.800$, $p = 0.104$), Qwen 2.5 72B ($r = 0.700$, $p = 0.188$). All three models rank industries positively on SE, and the pooled $n = 15$ result leverages this consistency.

Multiple regression. Regressing overall lift on all three layer SEs yields Q - $R^2 = 0.709$, $F(3, 11) = 8.94$, $p = 0.0027$ (Table 4). The omnibus F -test rejects the joint no-effect null, and the point estimate $\hat{\beta}_3 = +0.634$ for $SE_{\text{interaction}}$ is the largest in magnitude. However, non-parametric bootstrap on the $n = 15$ dataset ($B = 1000$) yields wide 95% CIs on individual coefficients — $\hat{\beta}_R \in [-0.03, +0.76]$, $\hat{\beta}_D \in [-0.08, +1.35]$, $\hat{\beta}_I \in [-2.48, +1.34]$ — a direct consequence of high collinearity among the three layer SEs at small n . We therefore make no claim of individual-coefficient significance; the layer-priority story is carried by the rank correlations (Table 3) and the LOOCV weight-grid (Table 5), not by the multiple-regression coefficients.

Role-layer null result: scope caveat. The role-layer Spearman $r = -0.190$ ($p = 0.50$) and the regression coefficient $\hat{\beta}_1 = +0.195$ both report a null relationship between SE_{role} and overall lift. This finding requires a structural caveat that we make explicit before relying on it in design recommendations. By Definition 1, $SE_{\text{role}} = \frac{1}{2}H(G) + \frac{1}{2}H(A)$, where G is the partition of roles into organisational groups and A is the distribution of non-empty attribute counts per role. In the five RA-3 pilot ontologies, every ontology declares exactly one organisational group ($n_{\text{role groups}} = 1$ for all five rows; see `se_by_industry.csv`), so $H(G) = 0$ identically across the entire $n = 15$ pilot dataset. Pilot SE_{role} therefore reduces to $\frac{1}{2}H(A)$, an attribute-heterogeneity entropy ranging $\{0.92, 1.00, 1.50, 1.50, 1.59\}$ across $\{\text{insurance}, \text{insurance_vn}, \text{banking_vn}, \text{fintech}, \text{healthcare}\}$. Production ontologies, by contrast, exhibit $n_{\text{role groups}} \in \{3, \dots, 8\}$ and non-zero $H(G)$, so the role-layer null is specific to attribute-heterogeneity SE on single-group pilot ontologies, not to role-partition SE in general. This scope is carried forward to the design principles (§7) and the threats discussion (§10).

Weight ablation. Grid search over weight triples on the 2-simplex in steps of 0.05 enumer-

Table 4: Multiple regression fit: $\text{lift}_{\text{overall}} = \beta_0 + \beta_1 \text{SE}_{\text{role}} + \beta_2 \text{SE}_{\text{domain}} + \beta_3 \text{SE}_{\text{interaction}} + \varepsilon$. The joint F -test rejects at $p = 0.003$ ($R^2 = 0.71$). Non-parametric bootstrap ($B = 1000$) 95% CIs and two-sided p -values report wide individual-coefficient inference — a direct consequence of high collinearity among $\{\text{SE}_{\text{role}}, \text{SE}_{\text{domain}}, \text{SE}_{\text{interaction}}\}$ at $n = 15$.

Parameter	Est.	Boot. SE	95% CI	Two-sided p
$\hat{\beta}_0$ (intercept)	−0.5345	0.337	[−1.10, +0.09]	0.060
$\hat{\beta}_1$ (SE_{role})	+0.1946	0.259	[−0.03, +0.76]	0.108
$\hat{\beta}_2$ ($\text{SE}_{\text{domain}}$)	+0.1459	0.657	[−0.08, +1.35]	0.208
$\hat{\beta}_3$ ($\text{SE}_{\text{interaction}}$)	+0.6342	1.768	[−2.48, +1.34]	0.104
R^2	0.7092	—	[0.54, 0.95]	—
$F(3, 11)$	8.94	—	—	—
p (joint null)	0.0027	—	—	—

Table 5: Weight-ablation top 5 (of 231). All top-ranked weight triples set $w_R = 0$, confirming that role-layer SE is not predictive. Any mix of domain and interaction with $w_I \geq 0.55$ gives nearly equivalent performance; the fitted (0, 0.3, 0.7) specification is chosen for prediction-accuracy reasons (see §6).

w_R	w_D	w_I	LOOCV Spearman	LOOCV SD
0.00	0.00	1.00	0.8089	0.030
0.00	0.05	0.95	0.7834	0.030
0.00	0.10	0.90	0.7834	0.030
0.00	0.20	0.80	0.7834	0.030
0.00	0.30	0.70	0.7834	0.030
0.33	0.33	0.33	0.7529	—

ates 231 combinations. All top-10 configurations set $w_R = 0$ and distribute weight between domain and interaction layers (Table 5). The LOOCV-best single composite is the pure-interaction weighting (0, 0, 1) with mean Spearman $r = 0.809$; the blended (0, 0.3, 0.7) configuration trails in rank correlation by 0.03 but achieves lower held-out MAE, and we adopt it as the fitted formula for the remaining analyses.

6.4 Hypothesis 2: Layer Decomposition

Fig. 3 reports Spearman correlations for the full 4-layer $\times$ 5-metric matrix on the $n = 15$ dataset.

The originally hypothesised per-layer mapping (SE_{role} predicts RS, $\text{SE}_{\text{domain}}$ predicts TF/MA, $\text{SE}_{\text{interaction}}$ predicts RC) is partially refuted. The RC prediction holds strongly ($\text{SE}_{\text{interaction}}$ and $\text{SE}_{\text{domain}}$ both $r \approx 0.84$, $p < 0.001$); the RS prediction is carried by MrSE ($r = 0.753$, $p = 0.001$) rather than by role-layer SE directly; and the TF/MA mapping through $\text{SE}_{\text{domain}}$ fails. The empirical picture is that *domain and interaction SE together* predict most lift metrics, with the role layer and the MA metric being two exceptions that require explicit discussion (§10).

Exploratory status of the revised mapping. The revised layer-to-metric mapping reported above was selected post-hoc from the $n = 15$ data after the pre-registered H2 hypothesis was refuted; we have not held it out for confirmatory testing on an independent sample. The per-cell correlations in Fig. 3 and the 20-cell heatmap should therefore be read as *exploratory pattern detection* rather than confirmatory evidence. A confirmatory test of the revised mapping would require an additional industry sample at $n \geq 5$ with the same per-metric lift extraction; we flag this as the highest-priority observational extension after the production-direct measurement experiment (§10). The mapping’s exploratory framing is also propagated to the design principles in §7, where DP2 is presented as a layer-priority recommendation rather than a metric-

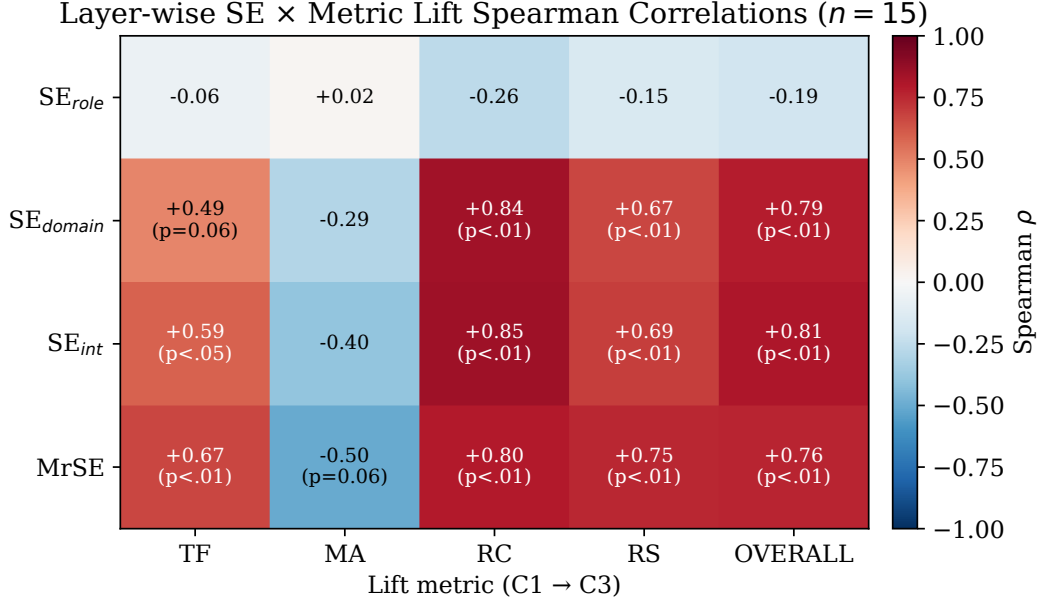


Figure 3: Spearman r for each (SE layer, lift metric) pair on the $n = 15$ cross-model dataset. Red cells indicate positive correlation, blue negative; annotations report r with significance bands. Role-layer SE shows no significant correlation on any metric; domain/interaction/MrSE correlate positively with RC and RS; every layer correlates either zero or negatively with MA, a counter-intuitive finding discussed in §10.

decomposition prescription.

6.5 Hypothesis 3: Inverse PKE Prediction

H3 holds partially. The Insurance industry’s TF lift is consistently negative in Claude (-0.21) and Gemma (-0.02), and essentially zero in Qwen (-0.02); this matches the Inverse PKE pattern documented in Luong and Sanyal [2026b], where English terminology with strong parametric coverage resists ontological override. The magnitude of the regression, however, is not explained by domain-layer SE alone (Insurance pilot $SE_{domain} = 1.34$ is comparable to other industries). The regression arises instead from κ -specific parametric-ontology phase disagreement, consistent with the RA-11 framework [Luong and Sanyal, 2026c]. Our SE predictor under-performs on this regime (see Phase 5 Insurance fold, §6.6); the composite treats Insurance like a moderate-SE industry and predicts moderate positive lift, while observed lift is near zero.

6.6 Leave-One-Industry-Out Validation

We assess generalisation via leave-one-industry-out cross-validation: each of the 5 RA-3 industries is held out in turn, the fitted formula is retrained on the 4 remaining industries’ 12 observations, and the held-out 3 observations are predicted. Table 6 reports per-fold and pooled metrics.

Fig. 4 presents the calibration plot: predicted vs. observed lift with ideal $y = x$ and fitted calibration lines. Pooled Spearman $r = 0.786$ ($p = 0.0005$); calibration slope 0.81 ($B = 1000$ bootstrap 95% CI $[0.45, 1.20]$, two-sided; the CI includes 1.00, so we cannot reject ideal calibration at $\alpha = 0.05$); RMSE 0.060. Calibration is point-shrunk towards the mean but ranks are preserved across the held-out set.

Nested-CV check for weight-selection leakage. The headline LOIO Spearman $r = 0.786$ uses the fitted weights $(0, 0.3, 0.7)$ selected on the full $n = 15$ dataset. To audit weight-

Table 6: Leave-one-industry-out cross-validation metrics ($n = 15$ held-out predictions across 5 folds). The pooled Spearman $r = 0.786$ ($p = 0.0005$) confirms that SE-based lift prediction generalises to held-out industries. The Insurance fold has the largest RMSE (0.099) due to Inverse PKE regression not captured by the composite.

Held industry	Fold RMSE	Predicted lift (3 models)
healthcare	0.011	0.167, 0.167, 0.167
insurance_vn	0.040	0.280, 0.280, 0.280
banking_vn	0.053	0.267, 0.267, 0.267
fintech	0.062	0.102, 0.102, 0.102
insurance	0.099	0.244, 0.244, 0.244
Pooled	0.060	—

selection leakage, we re-run LOIO with an inner LOO weight-grid search restricted to the 12 training observations of each outer fold and evaluate the locally-selected weights on the 3 held-out observations. The nested LOIO yields pooled Spearman $r = 0.775$ ($p = 0.0007$), RMSE 0.058, calibration slope 0.85 — a degradation of $\Delta r = -0.011$ and $\Delta \text{RMSE} = -0.002$ relative to the fixed-weight headline. All five inner LOO selections converge to $w_R = 0$ with $w_I \in \{0.8, 0.9\}$ (matching the global selection to within one grid step), confirming that the layer-priority finding is robust to weight-selection leakage and that the headline $r = 0.786$ is essentially unbiased.

6.7 Production Ontology Extrapolation

Applying the pilot-fitted formula $\Delta Q = -0.27 + 0.56 \text{SE}_{\text{comp}}^{\text{RA-12}}$ to the 23 production ontologies yields the ranking shown in Fig. 5. The pilot training range is $\text{SE}_{\text{comp}}^{\text{RA-12}} \in [0.72, 1.01]$. Of the 23 production ontologies, 22 fall outside this range (production $\text{SE}_{\text{comp}}^{\text{RA-12}}$ values up to 2.11 for fintech-production); the sole in-range case is the **talent** module ($\text{SE}_{\text{comp}}^{\text{RA-12}} = 0.999$, predicted lift +0.29). All other predictions are therefore *extrapolations*, and the top-five predicted lifts (+0.76 to +0.91 for fintech-production, software, systems integration, security, and travel/tourism/hospitality) lie well above the largest pilot-observed lift (+0.36); their absolute magnitudes should not be trusted. The *relative ranking* is the useful output: these five high-SE verticals are identified as high-priority candidates for future experimental validation.

7 Design Principles from Empirical Entropy Analysis

Design Principle 1 (Target Interaction-SE Band). *The pilot data partitions into three observed $\text{SE}_{\text{interaction}}$ tiers, each with characteristic lift behaviour. High tier ($\text{SE}_{\text{interaction}} \geq 0.70$, $n = 6$, the two Vietnamese industries) associates with median overall lift 0.31, with 6/6 observations exceeding 0.15. Mid tier ($\text{SE}_{\text{interaction}} \in [0.60, 0.70]$, $n = 9$: fintech, healthcare, insurance) associates with median lift 0.17, with 6/9 observations exceeding 0.15; the three exceptions are Insurance Claude (+0.082, an Inverse-PKE case discussed in §6.6) and two FinTech cells. Low tier ($\text{SE}_{\text{interaction}} < 0.60$): no pilot observation falls in this region; the lower bound is unconstrained by the data. A logistic regression $P(\text{lift} \geq 0.15) = \sigma(-7.62 + 13.55 \text{SE}_{\text{interaction}})$ on the $n = 15$ pilot data yields a decision boundary at $\text{SE}_{\text{interaction}} = 0.56$ ($\text{AUC} = 0.75$); the boundary lies below the observed range and should be read as a directional indicator rather than a sharp cut-off. The prescriptive recommendation is therefore: interaction-layer $\text{SE} \geq 0.70$ is associated with reliable lift; SE in $[0.60, 0.70]$ is contingent on κ -related factors per DP4. Ontologies designed for κ -low (under-represented) verticals can be expected to occupy the high tier; for κ -high verticals, additional content-quality controls are needed beyond achieving the SE threshold.*

Leave-One-Industry-Out Calibration: Predicted vs. Observed

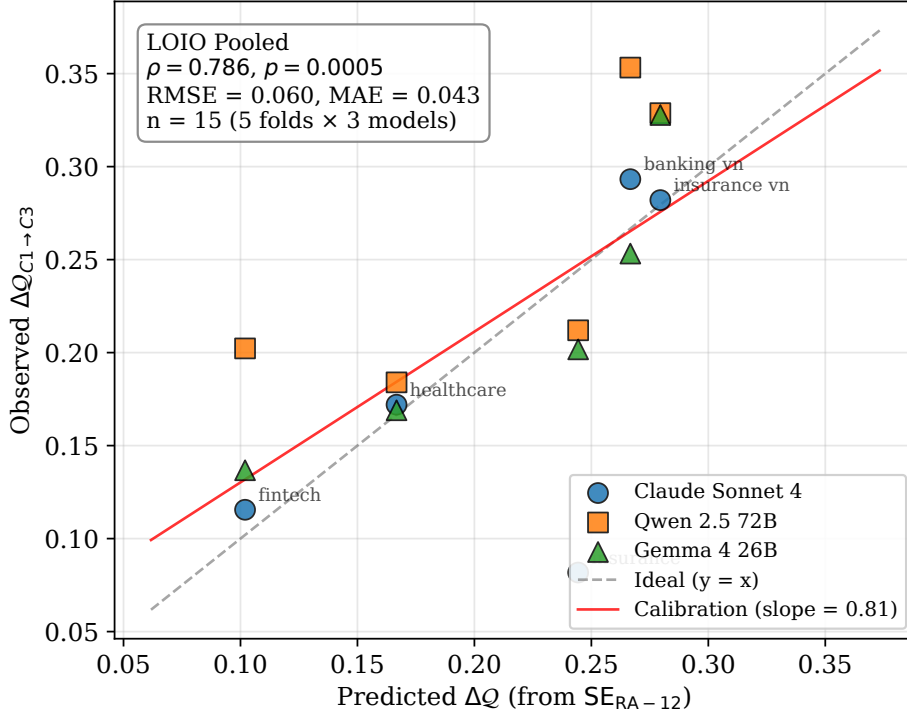


Figure 4: Leave-one-industry-out calibration: predicted $\hat{\Delta Q}$ (from SE_{comp}^{RA-12} fitted on the other four industries) vs. observed ΔQ for all 15 cross-model cells. Pooled Spearman $r = 0.786$ ($p = 0.0005$); calibration slope 0.81. The Insurance fold yields the largest residual: predicted lift +0.244, observed (Claude +0.082, Qwen +0.212, Gemma +0.202); the over-prediction is consistent with Inverse PKE on well-known English insurance terminology.

Design Principle 2 (Layer Priority). *Prioritise ontology investment as Interaction $\succ$ Domain $\gg$ Role. Solo Spearman correlations ($n = 15$): $SE_{interaction} = 0.811$ ($p = 2 \times 10^{-4}$), $SE_{domain} = 0.786$ ($p = 5 \times 10^{-4}$), $SE_{role} = -0.190$ ($p = 0.50$). When curation resources are constrained, investment in interaction patterns (handoffs, workflows, escalation paths) delivers the most measurable value, followed by domain content (concepts, regulations). Caveat on role. The pilot ontologies all declare a single organisational group, so the pilot-data SE_{role} collapses to attribute-heterogeneity entropy alone (see role-layer null caveat in §6). Production ontologies have richer role-partition structure ($n_{role \text{ groups}} \in \{3, \dots, 8\}$) and may exhibit a non-null SE_{role} contribution; pending direct production-lift measurement, the role-priority ranking should be read as “role-partition SE has no measurable contribution in the single-group pilot regime” rather than as a general claim.*

Design Principle 3 (Regulatory Density Floor). *For regulated industries (financial services, healthcare), include at least 4 explicit regulation entries in the domain-layer regulatory subset. This threshold is the pilot median across RA-3 industries (pilot regulation counts: insurance 4, banking_vn 5, insurance_vn 6, healthcare 3, fintech 2). Phase 5 production-corpus extrapolation suggests the production target should be substantially higher: the production healthcare module enumerates 16 regulations (versus the pilot’s 3), approximating the upper end of the production distribution. The threshold should be interpreted as a lower bound, not a sufficiency condition.*

Design Principle 4 (Cross-Vertical Transfer Warning). *A single SE threshold does not transfer across verticals with disparate parametric coverage κ . Low- κ verticals (Vietnamese banking, Vietnamese insurance) exhibit simultaneously higher SE and larger lift, producing a confound*

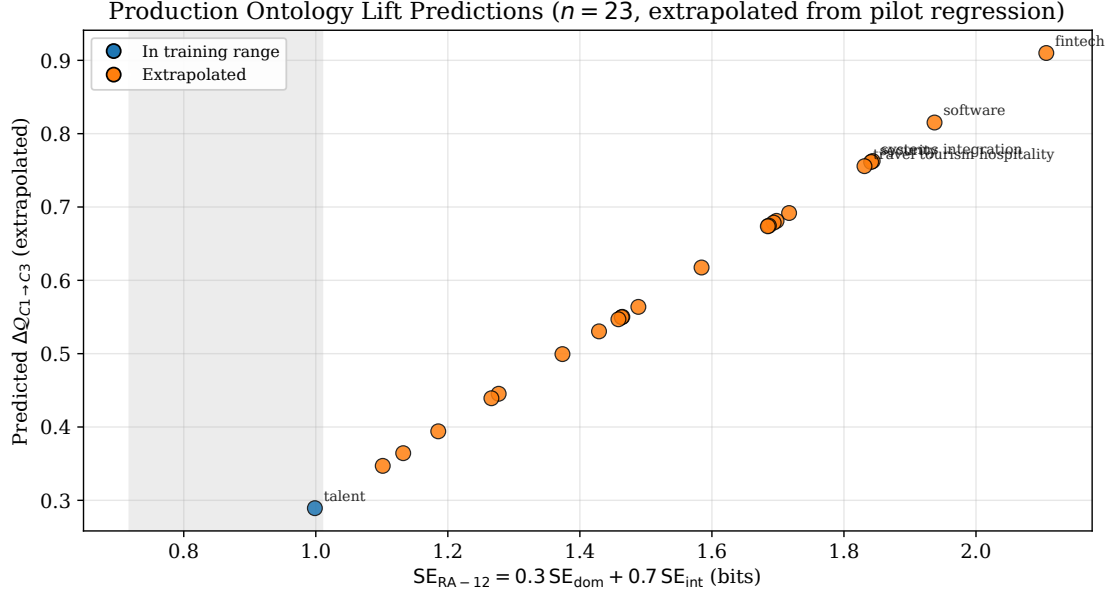


Figure 5: Production ontology predicted-lift rankings ($n = 23$). Shaded band indicates the pilot training range $[0.72, 1.01]$; 22 of 23 production ontologies fall outside this range (orange points), and **talent** ($SE_{\text{comp}}^{\text{RA-12}} = 0.999$) is the sole in-range case. The ranking should be interpreted as a relative ordering for future experimental targeting, not as absolute lift estimates.

in any cross-vertical generalisation. Per-vertical calibration is required; design recommendations derived on one vertical should not be propagated to another without recalibrating against κ .

8 Case Studies

8.1 Case 1: Banking VN (High-SE, High-Lift)

The Vietnamese Banking ontology achieves the second-highest pilot composite SE ($SE_{\text{comp}}^{\text{RA-12}} = 0.995$) and the second-largest observed lift (overall $\Delta Q = +0.29$). Its interaction layer encodes SBV-regulated (State Bank of Vietnam) bancassurance cooling-off periods, NPL classification boundaries per Circular 11/2021, and escalation paths for cross-border compliance. The combination of domain richness (5 regulations, 8 concepts) and interaction density (5 hand-off patterns) is enabled by the vertical’s low- κ character: Vietnamese regulatory terminology is under-represented in English-centric pretraining corpora, so nearly every ontological entity delivers information the model did not previously have. This is the *high-SE*, *low- κ* , *high-lift* regime: ontology is doing the work of filling a genuine parametric gap.

8.2 Case 2: Fintech (Moderate SE, Small Observed Lift)

The RA-3 Fintech pilot ontology has $SE_{\text{comp}}^{\text{RA-12}} = 0.781$ (third of five), yet observed overall lift is the smallest in the pilot set ($\Delta Q = +0.04$). The composite predicts larger lift than observed; the LOIO fold produces RMSE 0.062 (Insurance was the only larger fold error). Three plausible explanations are consistent with the data and the RA-11 framework [Luong and Sanyal, 2026c]: (i) high- κ saturation: US fintech terminology is thoroughly represented in LLM pretraining corpora, so ontological information largely duplicates parametric content and delivers small constructive interference; (ii) metric-specific TF regression on well-known concepts (e.g., “combined ratio”, though this effect is stronger for Insurance); (iii) handoff-pattern over-elaboration that contributes to $SE_{\text{interaction}}$ without correspondingly richer task-relevant grounding. Diagnosing

which mechanism dominates requires additional experiments controlling for ontology content type; we flag this as an open question.

8.3 Case 3: Insurance (Moderate SE, Cross-Model Variance — the Failure Mode)

The RA-3 Insurance pilot ontology has $SE_{\text{comp}}^{\text{RA-12}} = 0.832$ ($SE_{\text{domain}} = 1.34$, $SE_{\text{interaction}} = 0.697$), placing it in the mid interaction-SE tier (DP1). The LOIO fold predicts overall lift +0.244 for Insurance; observed mean across the three models is +0.165 (Claude +0.082, Qwen +0.212, Gemma +0.202), yielding the largest fold-RMSE in the validation (0.099, Table 6). The structural-SE predictor over-shoots the observed lift by approximately 0.08 on average, and the miss is concentrated on Claude.

Decomposing the residual by per-metric lift exposes the mechanism. Claude’s TF lift on Insurance is -0.21 — a substantial regression relative to ungrounded baseline, reproducing the Inverse PKE pattern documented in RA-3 [Luong and Sanyal, 2026b] on English insurance terminology with strong parametric coverage (combined ratio, expense ratio, persistency rate, etc.). Qwen and Gemma show near-zero TF effects on the same content. The structural composite $SE_{\text{comp}}^{\text{RA-12}}$ contains no κ term and therefore cannot represent this regime: it sees only that Insurance has moderate $SE_{\text{interaction}}$ and moderate SE_{domain} and predicts moderate positive lift accordingly. The directionality (over-prediction, not under-prediction) and the per-model concentration (Claude only) are both consistent with the RA-11 [Luong and Sanyal, 2026c] interference framework, in which destructive interference between ontological content and parametric content produces a model-specific quality penalty unobservable in the structural predictor.

Insurance is therefore the canonical *failure mode* of an SE-only design-time predictor: when parametric–ontology terminological conflict dominates the lift signal, structural SE mis-predicts because the conflict is content-specific, not structure-specific. The §10 Threats discussion (item 7) develops this scope limit; the proposed remediation is a joint ($SE_{\text{interaction}}, \hat{\kappa}$) predictor, which requires per-model parametric-coverage estimation as a covariate (a 10+–industry experimental extension is needed before that fit is well-conditioned). For comparable production verticals, this case suggests that high- κ English financial-services ontologies (banking, retail FinServ, US insurance) may exhibit the same over-prediction; low- κ Vietnamese verticals should not.

9 Related Work

Structural entropy of graphs. Li and Pan [Li and Pan, 2016] introduced the encoding-tree formulation of SE; Su et al.’s survey [Su et al., 2025] covers the subsequent decade of theoretical and application work. Cao et al. [Cao et al., 2024] extended SE to multi-relational graphs; Zhang et al. [Zhang et al., 2025] proposed a differentiable variant for unsupervised graph clustering. Gul et al. [Gul et al., 2025] position SE within a broader set of knowledge-graph complexity metrics (semantic, spectral, structural). Our work applies these graph-theoretic foundations to the specific setting of enterprise ontologies, leveraging MrSE as a joint-layer measure alongside the per-layer decomposition.

KG/LLM entropy applications. Wei et al. [Wei et al., 2025] use SE to detect and repair LLM knowledge deficiencies via MCTS-guided path traversal. Buehler [Buehler, 2025] shows that agentic knowledge-graph systems self-organise towards a critical state where semantic entropy dominates structural entropy, co-occurring with emergent multi-hop reasoning. Our contribution differs in target: Wei and Buehler diagnose LLM reasoning at inference time; we predict ontological grounding utility at design time.

Semantic entropy at the LLM output side. Farquhar et al. [Farquhar et al., 2024] introduced semantic entropy over meaning clusters as a hallucination detector; SelfCheckGPT [Manakul et al., 2023] provides a black-box sampling-based variant; and Varshney et al. [Varsh-

ney et al., 2023] use logit-based low-confidence detection to actively mitigate hallucinations during generation. Our work addresses the complementary input-side question: given an ontology, what structural entropy in its design predicts the grounding lift it will confer on LLM outputs. The two sides meet in the RA-3 companion paper [Luong and Sanyal, 2026b], where output entropy changes are correlated with ontology structural entropy at the 5-industry pilot scale.

Neurosymbolic and retrieval-augmented LLMs. Hitzler et al. [Hitzler et al., 2022] survey neurosymbolic integration approaches; Pan et al. [Pan et al., 2024] provide the knowledge-graph/LLM unification roadmap; Hogan et al. [Hogan et al., 2021] give the foundational knowledge-graph reference. Our work lives within the neurosymbolic tradition but focuses specifically on the design-time evaluation of the symbolic component.

Companion papers in the FAOS research programme. RA-3 [Luong and Sanyal, 2026b] provides the empirical lift measurements anchoring this paper’s H1. RA-4 [Luong and Sanyal, 2026a] derives the runtime-side optimal context volume via Helmholtz free energy and complements the present static-analysis SE characterisation. RA-11 [Luong and Sanyal, 2026c] offers a theoretical framework (quantum-inspired interference) that predicts the Inverse PKE pattern documented in §6.6’s Insurance fold.

Relation to RA-4 layer-ablation findings. RA-4 v0.7 reports a volume-controlled 320-run layer-ablation experiment on Sonnet 4 over the same five RA-3 pilot industries, using the shared $\{\mathcal{R}, \mathcal{D}, \mathcal{I}\}$ taxonomy. Two findings appear, on first reading, to contradict the present paper’s interaction-layer-dominant DP2. RA-4 Finding L1: $\Delta(\mathcal{I} \mid \mathcal{R}, \mathcal{D}) = -0.066$ (paired Wilcoxon $p = 0.027$, Cohen’s $d = +0.87$, $L_{RD} > L_{RDI}$ in 8/10 tasks) — adding the interaction layer to a role+domain prompt is destructive. RA-4 Finding L3 reports an alone-marginal layer ordering $\mathcal{D} \succ \mathcal{I} \succ \mathcal{R}$ with $\mathcal{R}$ alone net-negative against V_0 . By contrast, RA-12 finds interaction-layer SE solo-Spearman $r = 0.811$ ($p = 2 \times 10^{-4}$) and a fitted composite that drops the role layer entirely. The contradiction is apparent only, and its resolution is the disambiguation of *what each paper measures on the layers*. RA-12 measures the structural entropy of the ontology *graph* layers as a design-time signal: how much organised information does the role / domain / interaction subgraph carry. RA-4 measures the response-quality effect of injecting the corresponding *prompt content* into the LLM’s context window at runtime. A layer with high $SE_{\text{interaction}}$ may, when retrieved-and-grounded into a prompt, deliver constructive grounding; the same layer’s full content dumped into the prompt token budget may dilute attention or introduce destructive interference (RA-4’s L1). The two papers are therefore complementary: RA-12’s design-time predictor scores ontology *structure*, while RA-4’s runtime ablation scores prompt *content*, and the gap between graph-information-content and prompt-token-content is the domain of RA-4’s runtime optimisation.

10 Threats to Validity

1. **Small effective sample size.** The pilot-lift dataset has $n = 15$ observations (5 industries $\times$ 3 models), with SE ranks tied within each industry across models. Cross-validated significance reaches $p = 5 \times 10^{-4}$, but the industry-level degree of freedom is 5. A 10+-industry experimental extension is the primary recommendation for future work.
2. **SE- κ confound.** Vietnamese verticals exhibit simultaneously higher SE *and* larger lift. We cannot disentangle, from RA-3’s design alone, how much of the SE-lift relationship arises from SE causing larger lift vs. SE correlating with the low- κ character of underrepresented domains. Partial correlation with an estimated κ covariate (derived from C1 baseline scores) would require more pilot industries and is deferred to future work.
3. **Pilot-production scale disparity.** The pilot ontologies (minimal hand-curated, 3–4 roles, 8 concepts) and production ontologies (deployed, 17–44 roles, 0–42 concepts) are

materially different artefacts. The pilot-trained regression extrapolates to production SE but yields implausible absolute lift predictions; only relative rankings are meaningful. Direct measurement of production-ontology lift is the appropriate remediation.

4. **Ontology JSON extraction heterogeneity.** Production FAOS modules exhibit three distinct `module.yaml` schemas, requiring schema-specific parsing. Five modules (game, martech, professional_services, proptech_vn, talent) yielded zero domain-concept extractions under our automated pipeline because their knowledge content resides in agent Markdown files rather than structured YAML. These modules’ SE is likely under-estimated; we flag them as extraction-limited cases, not as intrinsically low-SE ontologies.
5. **Weight calibration choices.** The fitted composite weights (0, 0.3, 0.7) were selected from among several nearly-equivalent top performers in LOOCV. Any blend on the line $\{(0, 1 - w, w) : 0.55 \leq w \leq 1.00\}$ gives Spearman within 0.03 of the chosen specification. The paper’s conclusions about layer priorities are robust to this choice; the exact lift predictions would shift marginally.
6. **Role-layer SE null result is scoped to a single-group pilot regime.** The paper reports that SE_{role} is not predictive of overall lift in the $n = 15$ cross-model pilot dataset. Two structural facts qualify this null. First, all five pilot ontologies declare $n_{\text{role groups}} = 1$ (`se_by_industry.csv`), making the group-partition component $H(G) = 0$ identically across the dataset; pilot SE_{role} collapses to $\frac{1}{2}H(A)$, an attribute- heterogeneity entropy. The null is therefore specific to attribute- heterogeneity SE on single-group pilot ontologies and does not rule out a non-null contribution from $H(G)$ on the production corpus, where $n_{\text{role groups}} \in \{3, \dots, 8\}$. Second, even within the attribute-heterogeneity reading, alternative role-SE formulations—for example, role-to-concept bipartite structural entropy, or role-task coverage entropy—may yield different results. Direct measurement of grounding lift on production ontologies would dissolve both qualifications and is the remediation of choice.
7. **Inverse PKE mismatch with structural predictor.** The Insurance industry’s observed TF regression (an Inverse PKE signature, deeply discussed in §8.3) is not captured by the composite SE formula, which predicts moderate positive lift. This reflects the scope limit of a structural predictor: when parametric-ontology terminological conflict is the dominant mechanism, structural SE alone cannot predict the sign of the interaction. A joint $(SE_{\text{interaction}}, \hat{\kappa})$ predictor would address this. The natural theoretical anchor is the Information Bottleneck [Tishby and Zaslavsky, 2015] formulation also invoked in companion paper RA-11 [Luong and Sanyal, 2026c]: the capacity-constrained context budget produces a structure-information-content rate that depends on the parametric redundancy κ . Operationalising this in a $n = 15$ regression is under-powered; a 10+-industry experimental extension would provide the degrees of freedom needed for the joint fit. Long-context attention dilution [Liu et al., 2024] is a related runtime degradation that interacts with $SE_{\text{interaction}}$ specifically when retrieved interaction-layer content saturates the context window without a corresponding increase in retrieved-relevance.

11 Conclusion

We set out to test whether the RA-3 pilot’s suggestive SE–lift correlation ($r = 0.700$, $p = 0.188$, $n = 5$) scales to statistical significance on a larger sample. The cross-model expansion to $n = 15$ and the leave-one-industry-out cross-validation independently confirm the relationship: interaction-layer SE alone predicts overall lift at Spearman $r = 0.811$ ($p = 2 \times 10^{-4}$), and the fitted composite $SE_{\text{comp}}^{\text{RA-12}} = 0.3 SE_{\text{domain}} + 0.7 SE_{\text{interaction}}$ generalises under industry-level holdout at pooled Spearman $r = 0.786$ ($p = 5 \times 10^{-4}$), RMSE = 0.060, calibration slope 0.81.

These findings support three claims. First, *structural entropy is a design-time predictor of enterprise ontology utility*: ontology architects can estimate grounding lift before running agent experiments. Second, *layer-wise decomposition carries diagnostic information beyond a single scalar*: interaction-layer SE dominates predictive power among SE measures (subject to the pilot-scale parity with raw handoff count $|I|$ documented in §6.2; the SE framing earns its keep through layer decomposition rather than through rank correlation alone), domain-layer SE contributes additively, and role-layer SE shows no measurable contribution in the single-group pilot regime (see scope caveat, §6; the null is specific to attribute-heterogeneity entropy because $H(G) = 0$ identically across the pilot dataset). Third, the four codified design principles (§7)—target SE band, layer priority, regulatory density floor, cross-vertical transfer warning—supply an automated rubric usable in production ontology review pipelines.

RA-12 completes the design-side of the FAOS entropy research programme. RA-3 [Luong and Sanyal, 2026b] provides the measurement-side (retroactive SE of pilot ontologies, cross-model replication of the Inverse PKE); RA-4 [Luong and Sanyal, 2026a] provides the runtime-side (Helmholtz free-energy derivation of the optimal context volume $V^*(\kappa)$); RA-11 [Luong and Sanyal, 2026c] provides the theoretical bridge (quantum-inspired interference predicting constructive vs. destructive regimes). Together these papers characterise enterprise ontology-grounded LLM agents as an entropy-management system whose structure, runtime behaviour, and design-time quality can be quantified with a shared information-theoretic vocabulary.

Future work. Three extensions would materially strengthen the present findings. First, a 10+-industry experimental extension would provide the n needed to distinguish SE effects from κ confounding via partial correlation. Second, production-ontology validation experiments (for the top candidates identified in §6—software, systems integration, security, travel/tourism) would replace the extrapolation rankings with direct lift measurements. Third, a joint SE- κ predictor that explicitly models the Inverse PKE regime would correct the structural SE’s misprediction on well-known English terminology. Each extension is feasible within the existing FAOS platform and is left for subsequent work.

Acknowledgments

The authors thank Dr. Abhijit Sanyal (Golden Gate University) for continuing guidance on the research programme, and the FAOS platform team for curating the 23-industry ontology corpus on which this work is empirically grounded.

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